A SESSION OF THE 101ST KPPY WORKSHOP

Young KPPY

DATE
November 13, 2026 (Friday)

Talks 14:00–17:30, dinner 18:00

VENUE
B06, College of Natural Sciences
Kyungpook National University, Daegu

youngkppy.seminresearch.com

PROGRAM

14:00–14:45	**Hong-Jun Ge** University of Science and Technology of China On co-edge-regular graphs with four distinct eigenvalues
14:45–14:55	Break
14:55–15:40	**Hyungkee Yoo** Sunchon National University On Three- and Four-Page Presentations of Knots and Links
15:40–15:50	Break
15:50–16:35	**Sunyo Moon** Kumoh National Institute of Technology Edge completion of signed graphs with spectral integral variation
16:35–16:45	Break
16:45–17:30	**Jingyuan Liu** University of Science and Technology of China On signed graphs with smallest eigenvalue at least -3 and their lattices
18:00	Dinner

Each talk is 40 minutes plus 5 minutes for questions.

14:00–14:45

On co-edge-regular graphs with four distinct eigenvalues

Hong-Jun Ge University of Science and Technology of China

Joint work with Edwin R. van Dam and Jack H. Koolen

A connected k-regular graph on v vertices is called co-edge-regular with parameters (v, k, μ) if every pair of distinct nonadjacent vertices has exactly μ common neighbours. The λ-level of a graph G is the number of distinct values of $|N(x) \cap N(y)|$ as xy ranges over the edges of G.

Tan, Koolen, and Xia conjectured that, for a fixed smallest eigenvalue under suitable largeness and density assumptions, every connected co-edge-regular graph with exactly four distinct eigenvalues must be either a clique extension of a strongly regular graph or a grid graph. Both have λ-level 2.

In this talk, we present two constructions yielding infinite families of counterexamples. Every graph constructed is cospectral with a clique extension of a strongly regular graph, but has λ-level larger than 2.

The first construction starts from a regular partial linear space containing an affine or projective plane and applies Wang–Qiu–Hu (WQH) switching. For every order q for which such a plane exists, the construction yields infinitely many graphs with smallest eigenvalue $-2q - 1$ and λ-level at least $q + 2$. Thus, the λ-levels arising from this construction are unbounded as q varies.

The second construction is based on group-divisible orthogonal arrays and parallel classes arising from vector spaces over finite fields. The graphs in this family have smallest eigenvalue -5 and λ-level 3. More strikingly, the coherent closure of each graph is the full matrix algebra, so its coherent rank attains the maximum possible value v^2, where v denotes its number of vertices.

These constructions show that, even in the co-edge-regular setting, four eigenvalues does not preclude either unbounded local complexity or maximal coherent-algebraic complexity.

14:55–15:40

On Three- and Four-Page Presentations of Knots and Links

Hyungkee Yoo Sunchon National University

Page presentations provide combinatorial representations of knots and links by embedding them into a union of half-planes sharing a common boundary line. The corresponding page indices measure the minimum number of arcs required for such representations.

In this talk, I will introduce the history of three-page presentations and review some known results on the three-page index. I will then present a diagrammatic approach based on non-

separating independent sets in planar graphs, which yields the upper bound

$$\alpha_3(L) \leq 3c(L) - 1$$

for any non-split, nontrivial link L other than the Hopf link.

I will also introduce the four-page index and explain how spanning trees of checkerboard graphs lead to the improved bound

$$\alpha_4(L) \leq 2c(L).$$

Finally, I will discuss applications to ribbonlength and related combinatorial problems.

15:50–16:35

Edge completion of signed graphs with spectral integral variation

Sunyo Moon Kumoh National Institute of Technology

Joint work with Jungho Ahn and Cheolwon Heo

Spectral variation of two matrices is integral if their spectra differ by integer quantities. For two signed graphs, where one is obtained from the other by adding a new edge, we characterize when the spectral variation of their signed Laplacian matrices becomes integral. Furthermore, for every fixed signed complete graph, we fully characterize the class of signed graphs to which one can recursively add new edges keeping spectral integral variation to make the signed complete graph. This generalizes Kirkland's characterization for integrally completable graphs (Discrete Math., 2005).

16:45–17:30

On signed graphs with smallest eigenvalue at least -3 and their lattices

Jingyuan Liu University of Science and Technology of China

Joint work with Meng-Yue Cao, Jack H. Koolen, and Qianqian Yang

Graphs with smallest eigenvalue at least -2 can be represented by root lattices. In 2018, Koolen et al. gives a result on (unsigned) graphs with smallest eigenvalue at least -3 and their associated lattices. This talk presents an extension of this theory to signed graphs.

We prove that every connected signed graph with smallest eigenvalue greater than $-3-\varepsilon$ and sufficiently large minimum valency has smallest eigenvalue at least -3, and its associated lattice is a sublattice of $\mathbb{Z}^n \oplus E_8^m$ for some n, m.

Consequently, this also gives a result on rootless lattices generated by norm 3 vectors.